Linear Convergence of Gradient Descent on Two-Layer ReLU Networks via the Neural Tangent Kernel

1 Preliminaries

1.1 Notation

We use boldface for vectors and matrices. For a matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$, $\lambda_{\min}(\mathbf{A})$ and $\lambda_{\max}(\mathbf{A})$ denote the smallest and largest eigenvalues, and $\|\mathbf{A}\|_{\mathrm{op}}$ denotes its operator norm (largest singular value). The Euclidean norm of a vector is $\|\cdot\|_2$ and the Frobenius norm of a matrix is $\|\cdot\|_F$. We write $[n] \coloneqq \{1,\dots,n\}$. The indicator function of an event A is $\mathbf{1}[A] \in \{0,1\}$. The ReLU activation is $\sigma(z) \coloneqq \max(z,0)$; we adopt the convention $\sigma'(0) = 0$ for the (Lebesgue-null) non-differentiable point.

Throughout, $c, C, C_1, C_2, \dots$ denote universal positive constants whose values may change from line to line and depend only on quantities specified in each lemma's hypotheses. Problem-dependent constants carry their dependencies as subscripts when needed.

1.2 Setup and data

Fix a training set $\{(\mathbf{x}_i, y_i)\}_{i=1}^n \subset \mathbb{R}^d \times \mathbb{R}$. Denote the input matrix and target vector by $\mathbf{X} = [\mathbf{x}_1,\dots,\mathbf{x}_n] \in \mathbb{R}^{d\times n}$ and $\mathbf{y} = (y_1,\dots,y_n)^\top \in \mathbb{R}^n$. We consider a two-layer ReLU network with m hidden units,

$$f(\mathbf{W},\mathbf{a},\mathbf{x}) \;=\; \frac{1}{\sqrt{m}} \sum_{r=1}^{m} a_r\, \sigma\big(\mathbf{w}_r^\top \mathbf{x}\big), \tag{1}$$

where $\mathbf{W} = [\mathbf{w}_1,\dots,\mathbf{w}_m] \in \mathbb{R}^{d\times m}$ collects the first-layer weights and $\mathbf{a} = (a_1,\dots,a_m)^\top \in \mathbb{R}^m$ collects the second-layer weights. Throughout, the second-layer weights $\mathbf{a}$ are frozen at initialization; only $\mathbf{W}$ is trained, in line with [DZPS19]. We write $\mathbf{u}_i(\mathbf{W}) \coloneqq f(\mathbf{W},\mathbf{a},\mathbf{x}_i)$ for the network's output on input $\mathbf{x}_i$, $\mathbf{u}(\mathbf{W}) \coloneqq (\mathbf{u}_1(\mathbf{W}),\dots,\mathbf{u}_n(\mathbf{W}))^\top$, and define the training loss

$$L(\mathbf{W}) \;\coloneqq\; \tfrac{1}{2}\,\|\mathbf{u}(\mathbf{W}) - \mathbf{y}\|_2^2 \;=\; \tfrac{1}{2}\sum_{i=1}^{n} \big(\mathbf{u}_i(\mathbf{W}) - y_i\big)^2. \tag{2}$$

Gradient descent with step size $\eta > 0$ generates the sequence $\mathbf{W}(0), \mathbf{W}(1), \mathbf{W}(2), \dots$ via

$$\mathbf{W}(t+1) \;=\; \mathbf{W}(t) - \eta\, \nabla_{\mathbf{W}} L(\mathbf{W}(t)), \tag{3}$$

with the per-column gradient (suppressing the argument $\mathbf{W}(t)$)

$$\frac{\partial L(\mathbf{W})}{\partial \mathbf{w}_r} \;=\; \frac{a_r}{\sqrt{m}} \sum_{i=1}^{n} \big(\mathbf{u}_i(\mathbf{W}) - y_i\big)\, \mathbf{1}\big[\mathbf{w}_r^\top \mathbf{x}_i \ge 0\big]\, \mathbf{x}_i. \tag{4}$$

We use $\mathbf{u}(t) \coloneqq \mathbf{u}(\mathbf{W}(t))$ and $\mathbf{w}_r(t) \coloneqq \mathbf{w}_r$ at step t.

1.3 Assumptions

Assumption 1.1 (Normalized inputs and bounded labels)**.** For every $i \in [n]$, $\|\mathbf{x}_i\|_2 = 1$ and $|y_i| \le 1$.

Remark 1.2. Both conditions are standard in NTK analyses [DZPS19]; the label bound enters only through the constant $\|\mathbf{u}(0) - \mathbf{y}\|_2 = O(\sqrt{n})$ established in Lemma 2.5. Inputs may be rescaled to unit norm without loss of generality. The result extends to $\|\mathbf{x}_i\|_2 \le 1$ and $|y_i| \le C$ at the cost of a constant factor in m and η.

Assumption 1.3 (Random initialization)**.** The weights are initialized as

$$\mathbf{w}_r(0) \overset{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}_d), \qquad a_r \overset{\text{iid}}{\sim} \mathrm{Unif}\{-1, +1\},$$

independent across $r \in [m]$, and the two sequences $(\mathbf{w}_r(0))_r$ and $(a_r)_r$ are mutually independent.

Remark 1.4. This is the initialization scheme of [DZPS19]. Symmetry of a_r ensures $\mathbb{E}[\mathbf{u}_i(0)] = 0$ for every i, and the variance scaling $1/\sqrt{m}$ in Eq. (1) keeps $\mathbf{u}_i(0)$ stochastically bounded as $m \to \infty$.

1.4 The empirical and population Gram matrices

Definition 1.5 (Gram matrix at weights $\mathbf{W}$)**.** For weights $\mathbf{W} \in \mathbb{R}^{d\times m}$ and fixed sign vector $\mathbf{a} \in \{-1, +1\}^m$, the empirical Gram matrix $\mathbf{H}(\mathbf{W}) \in \mathbb{R}^{n\times n}$ has entries

$$\mathbf{H}(\mathbf{W})_{ij} \;\coloneqq\; \frac{1}{m} \sum_{r=1}^{m} \mathbf{x}_i^\top \mathbf{x}_j \, \mathbf{1}\big[\mathbf{w}_r^\top \mathbf{x}_i \ge 0\big] \, \mathbf{1}\big[\mathbf{w}_r^\top \mathbf{x}_j \ge 0\big] \qquad (i, j \in [n]). \tag{5}$$

Remark 1.6. $\mathbf{H}(\mathbf{W})$ is the Gram matrix of the per-sample gradient features $\{\nabla_{\mathbf{W}} \mathbf{u}_i(\mathbf{W})\}_{i=1}^n$: a direct computation from Eq. (4) gives $\langle \nabla_{\mathbf{W}} \mathbf{u}_i(\mathbf{W}), \nabla_{\mathbf{W}} \mathbf{u}_j(\mathbf{W})\rangle = a_r^2 m^{-1} \mathbf{x}_i^\top \mathbf{x}_j \mathbf{1}[\mathbf{w}_r^\top \mathbf{x}_i \ge 0] \mathbf{1}[\mathbf{w}_r^\top \mathbf{x}_j \ge 0] = \mathbf{H}(\mathbf{W})_{ij}$ since $a_r^2 = 1$.

Definition 1.7 (Population NTK Gram matrix)**.** The population NTK Gram matrix $\mathbf{H}^\infty \in \mathbb{R}^{n\times n}$ has entries

$$\mathbf{H}^\infty_{ij} \;\coloneqq\; \mathbb{E}_{\mathbf{w}\sim\mathcal{N}(\mathbf{0},\mathbf{I}_d)}\Big[\mathbf{x}_i^\top \mathbf{x}_j \, \mathbf{1}\big[\mathbf{w}^\top \mathbf{x}_i \ge 0\big] \, \mathbf{1}\big[\mathbf{w}^\top \mathbf{x}_j \ge 0\big]\Big] \qquad (i, j \in [n]). \tag{6}$$

Define $\lambda_0 \coloneqq \lambda_{\min}(\mathbf{H}^\infty)$.

Assumption 1.8 (Spectral assumption)**.** $\lambda_0 = \lambda_{\min}(\mathbf{H}^\infty) > 0$.

Remark 1.9. Assumption 1.8 is the single spectral hypothesis driving the whole proof. [DZPS19] show that $\lambda_0 > 0$ whenever no two inputs are parallel; i.e., $\mathbf{x}_i / \|\mathbf{x}_i\|_2 \ne \pm \mathbf{x}_j / \|\mathbf{x}_j\|_2$ for all $i \ne j$. The width and step-size requirements scale polynomially in $1/\lambda_0$.

1.5 External facts

Fact 1.10 (Hoeffding's inequality, bounded case)**.** Let $Z_1, \dots, Z_m$ be independent random variables with $\mathbb{E}Z_r = 0$ and $|Z_r| \le b$ a.s. Then for all $t \ge 0$,

$$\mathbb{P}\Big[\Big|\tfrac{1}{m}\textstyle\sum_{r=1}^m Z_r\Big| \ge t\Big] \;\le\; 2\exp\Big(-\tfrac{mt^2}{2b^2}\Big).$$

Fact 1.11 (Gaussian small-ball, one-dimensional)**.** Let $g \sim \mathcal{N}(0, 1)$. Then for all $R \ge 0$, $\mathbb{P}[\,|g| \le R\,] \le R\sqrt{2/\pi} \le R$.

2 Main result and lemma decomposition

2.1 Headline theorem

Theorem 2.1 (Linear convergence of gradient descent)**.** *Fix any $\delta \in (0,1)$. Suppose Assumptions 1.1, 1.3 and 1.8 hold and the width and step size satisfy*

$$m \;\geq\; C_1 \cdot \frac{n^6}{\lambda_0^4 \delta^3}, \qquad \eta \;=\; C_2 \cdot \frac{\lambda_0}{n^2}, \tag{7}$$

for sufficiently large $C_1 > 0$ and sufficiently small $C_2 > 0$ universal constants. Then with probability at least $1-\delta$ over the random initialization (Assumption 1.3), every iterate of gradient descent (Eq. (3)) satisfies

$$\|\mathbf{u}(t) - \mathbf{y}\|_2^2 \;\leq\; \left(1 - \tfrac{\eta\lambda_0}{2}\right)^t \|\mathbf{u}(0) - \mathbf{y}\|_2^2 \qquad \text{for all } t \geq 0. \tag{8}$$

Moreover, for every $r \in [m]$ and every $t \geq 0$,

$$\|\mathbf{w}_r(t) - \mathbf{w}_r(0)\|_2 \;\leq\; \frac{4\sqrt{n}\,\|\mathbf{u}(0) - \mathbf{y}\|_2}{\sqrt{m}\,\lambda_0} \;=:\; R. \tag{9}$$

Theorem 2.1 establishes that gradient descent on the squared loss converges to zero training loss at a linear rate $1 - \eta\lambda_0/2$, while every weight stays within a $1/\sqrt{m}$ ball of its initialization. The proof factors through three lemmas, stated next.

2.2 Lemma decomposition

Lemma 2.2 (Initial Gram concentration)**.** *Fix any $\delta \in (0,1)$. Suppose Assumptions 1.1 and 1.3 hold and*

$$m \;\geq\; \frac{C\, n^2 \log(n/\delta)}{\lambda_0^2},$$

for a sufficiently large universal constant $C > 0$. Then with probability at least $1 - \delta/3$ over the initialization,

$$\|\, \mathbf{H}(\mathbf{W}(0)) - \mathbf{H}^\infty \,\|_{\mathrm{op}} \;\leq\; \tfrac{\lambda_0}{4},$$

and in particular $\lambda_{\min}(\mathbf{H}(\mathbf{W}(0))) \geq \frac{3\lambda_0}{4}$.

Lemma 2.3 (Gram stability under bounded perturbation)**.** *Fix any $\delta \in (0,1)$ and let $R > 0$. Suppose Assumptions 1.1 and 1.3 hold and*

$$m \;\geq\; \frac{C\, n^2 \log(n/\delta)}{\lambda_0^2}, \qquad R \;\leq\; \frac{c\,\lambda_0}{n^2},$$

for a sufficiently large universal constant $C > 0$ and a sufficiently small universal constant $c > 0$. Then with probability at least $1-\delta/3$ over the initialization, the following holds for every $\mathbf{W} \in \mathbb{R}^{d\times m}$ with $\|\mathbf{w}_r - \mathbf{w}_r(0)\|_2 \leq R$ for all $r \in [m]$:

$$\|\, \mathbf{H}(\mathbf{W}) - \mathbf{H}(\mathbf{W}(0)) \,\|_{\mathrm{op}} \;\leq\; \tfrac{\lambda_0}{4}.$$

Lemma 2.4 (Linear convergence in a stay-in-ball regime)**.** *Let $R > 0$. Suppose that on a deterministic event the following hold simultaneously:*

- $\lambda_{\min}(\mathbf{H}(\mathbf{W}(s))) \geq \frac{\lambda_0}{2}$ *for every* $s \in \{0, 1, \ldots, t\}$*;*
- *For every* $i \in [n]$*, the set* $S_i := \{r \in [m] : A_{i,r}(R)$ *occurs*$\}$ *of Lemma 2.3's proof satisfies* $|S_i| \leq 2mR$ *(Eq. (16));*
- *For every* $r \in [m]$ *and every* $s \in \{0, 1, \ldots, t\}$*,* $\|\mathbf{w}_r(s) - \mathbf{w}_r(0)\|_2 \leq R$*;*
- $R \leq c\lambda_0/n^2$ *and* $\eta = C_2\lambda_0/n^2$ *for sufficiently small universal constants* $c, C_2 > 0$*.*

Then, on the same event, both

$$\|\mathbf{u}(s) - \mathbf{y}\|_2^2 \;\leq\; \left(1 - \tfrac{\eta\lambda_0}{2}\right)^s \|\mathbf{u}(0) - \mathbf{y}\|_2^2 \quad \textit{for every } s \in \{0, 1, \ldots, t+1\},$$

and, for every $r \in [m]$ *and every* $s \in \{0, 1, \ldots, t+1\}$*,*

$$\|\mathbf{w}_r(s) - \mathbf{w}_r(0)\|_2 \;\leq\; \frac{4\sqrt{n}\,\|\mathbf{u}(0) - \mathbf{y}\|_2}{\sqrt{m}\,\lambda_0}.$$

Lemma 2.5 (Initial residual bound)**.** *Suppose Assumptions 1.1 and 1.3 hold. Fix any* $\delta \in (0, 1)$*. Then with probability at least* $1 - \delta/3$ *over the initialization,*

$$\|\mathbf{u}(0) - \mathbf{y}\|_2^2 \;\leq\; C'\, n \log(n/\delta),$$

for a universal constant $C' > 0$*. In particular,* $\|\mathbf{u}(0) - \mathbf{y}\|_2 \leq \sqrt{C' n \log(n/\delta)}$*.*

Lemma 2.4 is the deterministic mechanism: as long as the Gram matrix's spectrum stays above $\lambda_0/2$, the squared residual contracts by a factor $1 - \eta\lambda_0/2$ per step, and the per-neuron movement is bounded by $4\sqrt{n}\,\|\mathbf{u}(0) - \mathbf{y}\|_2 / (\sqrt{m}\lambda_0)$, which is the radius R in Eq. (9). Lemmas 2.2 and 2.3 together guarantee, with high probability over initialization, that the Gram spectrum *does* stay above $\lambda_0/2$ throughout training, by a fixed-point / contradiction argument carried out in Section 6. Lemma 2.5 controls the magnitude of $\|\mathbf{u}(0) - \mathbf{y}\|_2$ so that the perturbation radius R implied by Lemma 2.4 fits inside the perturbation hypothesis of Lemma 2.3, which is where the width requirement $m \gtrsim n^6/(\lambda_0^4\delta^3)$ enters.

3 Proof of Lemma 2.2 and Lemma 2.5

3.1 Concentration of the initial Gram matrix

Proof of Lemma 2.2. The proof proceeds in two steps. First, we apply Hoeffding's inequality entrywise to obtain a uniform entrywise concentration $|\mathbf{H}(\mathbf{W}(0))_{ij} - \mathbf{H}^\infty_{ij}| \leq \frac{\lambda_0}{4n}$; second, we convert entrywise to operator-norm control via the elementary bound $\|\mathbf{A}\|_{\mathrm{op}} \leq n \max_{ij} |\mathbf{A}_{ij}|$.

Step 1: entrywise Hoeffding. Fix $i, j \in [n]$. By Eq. (5) and Eq. (6),

$$\mathbf{H}(\mathbf{W}(0))_{ij} - \mathbf{H}^\infty_{ij} \;=\; \frac{1}{m}\sum_{r=1}^{m} Z_r^{(ij)}, \qquad Z_r^{(ij)} \;:=\; \mathbf{x}_i^\top \mathbf{x}_j \big(\mathbf{1}[E_{i,r}]\mathbf{1}[E_{j,r}] - \mathbb{E}[\mathbf{1}[E_{i,r}]\mathbf{1}[E_{j,r}]]\big), \tag{10}$$

with $E_{i,r} := \{\mathbf{w}_r(0)^\top \mathbf{x}_i \geq 0\}$. Across $r \in [m]$, the variables $\{Z_r^{(ij)}\}_r$ are independent because $\{\mathbf{w}_r(0)\}_r$ is iid (Assumption 1.3). Each $Z_r^{(ij)}$ has mean zero and is bounded: $|\mathbf{1}[E_{i,r}]\mathbf{1}[E_{j,r}]| \leq 1$, $|\mathbf{x}_i^\top \mathbf{x}_j| \leq \|\mathbf{x}_i\|_2 \|\mathbf{x}_j\|_2 = 1$ by Cauchy-Schwarz and Assumption 1.1, so $|Z_r^{(ij)}| \leq 2$.

Apply Fact 1.10 with $b = 2$ at the level $t = \frac{\lambda_0}{4n}$:

$$\mathbb{P}\Big[\,\big|\mathbf{H}(\mathbf{W}(0))_{ij} - \mathbf{H}^{\infty}_{ij}\big| \;\geq\; \tfrac{\lambda_0}{4n}\,\Big] \;\leq\; 2\exp\Big(-\tfrac{m\lambda_0^2}{128\,n^2}\Big).$$

Union-bounding over the n^2 index pairs (i, j),

$$\mathbb{P}\Big[\max_{i,j\in[n]}\big|\mathbf{H}(\mathbf{W}(0))_{ij} - \mathbf{H}^{\infty}_{ij}\big| \;\geq\; \tfrac{\lambda_0}{4n}\,\Big] \;\leq\; 2n^2\exp\Big(-\tfrac{m\lambda_0^2}{128\,n^2}\Big). \tag{11}$$

The right-hand side is at most $\delta/3$ as soon as $m \geq \frac{128\,n^2}{\lambda_0^2}\log(6n^2/\delta)$, which is implied by the lemma's hypothesis $m \geq Cn^2\log(n/\delta)/\lambda_0^2$ for a sufficiently large universal C.

Step 2: entrywise to operator norm. On the event $\{\max_{i,j}|\mathbf{H}(\mathbf{W}(0))_{ij} - \mathbf{H}^{\infty}_{ij}| \leq \frac{\lambda_0}{4n}\}$, the elementary inequality $\|\mathbf{A}\|_{\mathrm{op}} \leq \sqrt{\|\mathbf{A}\|_1\,\|\mathbf{A}\|_\infty} \leq n\max_{i,j}|\mathbf{A}_{ij}|$ for $\mathbf{A} \in \mathbb{R}^{n\times n}$ (which itself follows from $\|\mathbf{A}\|_{\mathrm{op}}^2 \leq \|\mathbf{A}\|_F^2 \leq n^2\max_{ij}\mathbf{A}_{ij}^2$) gives

$$\|\mathbf{H}(\mathbf{W}(0)) - \mathbf{H}^{\infty}\|_{\mathrm{op}} \;\leq\; n\cdot\tfrac{\lambda_0}{4n} \;=\; \tfrac{\lambda_0}{4}.$$

By Weyl's inequality, $\lambda_{\min}(\mathbf{H}(\mathbf{W}(0))) \geq \lambda_{\min}(\mathbf{H}^{\infty}) - \|\mathbf{H}(\mathbf{W}(0)) - \mathbf{H}^{\infty}\|_{\mathrm{op}} \geq \lambda_0 - \frac{\lambda_0}{4} = \frac{3\lambda_0}{4}$. This completes the proof. □

3.2 Initial residual bound

Proof of Lemma 2.5. The strategy is to apply Hoeffding's inequality to each $\mathbf{u}_i(0)$ separately, then union-bound.

Fix $i \in [n]$. By Eq. (1) at $\mathbf{W} = \mathbf{W}(0)$,

$$\mathbf{u}_i(0) \;=\; \frac{1}{\sqrt{m}}\sum_{r=1}^{m} a_r\,\sigma(\mathbf{w}_r(0)^\top\mathbf{x}_i) \;=\; \frac{1}{\sqrt{m}}\sum_{r=1}^{m} Y_r^{(i)}, \qquad Y_r^{(i)} \;:=\; a_r\,\sigma(\mathbf{w}_r(0)^\top\mathbf{x}_i).$$

Condition on the first-layer draws $\mathbf{W}(0)$ and let $b_r := \sigma(\mathbf{w}_r(0)^\top\mathbf{x}_i) \geq 0$. Then $\mathbb{E}[Y_r^{(i)} \mid \mathbf{W}(0)] = b_r\mathbb{E}[a_r] = 0$ (since a_r is symmetric Rademacher and independent of $\mathbf{W}(0)$), $|Y_r^{(i)}| \leq b_r$ a.s., and the $Y_r^{(i)}$ are conditionally independent across r. Apply Fact 1.10 conditionally to $\mathbf{u}_i(0) = m^{-1/2}\sum_r Y_r^{(i)}$: the random variables $\tilde{Y}_r := Y_r^{(i)}$ are independent with $\mathbb{E}\tilde{Y}_r = 0$ and $|\tilde{Y}_r| \leq b_r$, so applying Fact 1.10 (in its un-normalized form $\mathbb{P}[|\sum_r \tilde{Y}_r| \geq u] \leq 2\exp(-u^2/(2\sum_r b_r^2))$) gives, for any $u \geq 0$,

$$\mathbb{P}\Big[\,m^{-1/2}\big|\textstyle\sum_r Y_r^{(i)}\big| \;\geq\; u \;\Big|\; \mathbf{W}(0)\Big] \;\leq\; 2\exp\Big(-\tfrac{mu^2}{2\sum_r b_r^2}\Big). \tag{12}$$

We now bound $\sum_r b_r^2$ in expectation and high probability. Each $b_r^2 = \sigma(\mathbf{w}_r(0)^\top\mathbf{x}_i)^2 \leq (\mathbf{w}_r(0)^\top\mathbf{x}_i)^2$, and $(\mathbf{w}_r(0)^\top\mathbf{x}_i)^2 \sim \chi_1^2$ by Assumption 1.3 and $\|\mathbf{x}_i\| = 1$. Hence $\sum_r b_r^2 \leq \sum_r(\mathbf{w}_r(0)^\top\mathbf{x}_i)^2 =: V$ with $V \sim \chi_m^2$, and $\mathbb{E}V = m$. A standard χ^2 tail (e.g. Lemma 1 of Laurent-Massart 2000, $\mathbb{P}[V \geq m + 2\sqrt{mx} + 2x] \leq e^{-x}$ for $x \geq 0$) at $x = cm$ for small $c > 0$ gives $\mathbb{P}[V \geq 2m] \leq \exp(-c'm)$ for some absolute $c' > 0$. On the event $\{\sum_r b_r^2 \leq 2m\}$ (which has probability $\geq 1 - \exp(-c'm)$), Eq. (12) simplifies to

$$\mathbb{P}\big[\,|\mathbf{u}_i(0)| \;\geq\; u \;\big|\; \mathbf{W}(0),\, \textstyle\sum_r b_r^2 \leq 2m\big] \;\leq\; 2\exp(-u^2/4).$$

Taking $u = \sqrt{4\log(12n/\delta)}$ yields $|\mathbf{u}_i(0)| \le u$ with conditional probability $\ge 1 - \delta/(6n)$; integrating over the conditioning,

$$\mathbb{P}\Big[\, |\mathbf{u}_i(0)| \;\ge\; 2\sqrt{\log(12n/\delta)} \,\Big] \;\le\; \delta/(6n) + \exp(-c'm).$$

For $m \ge Cn^2 \log(n/\delta)/\lambda_0^2$ as in the lemma hypothesis (a non-vacuous constraint, $m \ge \Omega(\log(1/\delta))$), the second term is dominated by the first. Union-bounding over $i \in [n]$,

$$\mathbb{P}\Big[\, \max_{i\in[n]} |\mathbf{u}_i(0)| \;\le\; 2\sqrt{\log(12n/\delta)} \,\Big] \;\ge\; 1 - \delta/3.$$

On this event, by $(a-b)^2 \le 2a^2 + 2b^2$ (AM-GM) and $|y_i| \le 1$ (Assumption 1.1),

$$\|\mathbf{u}(0) - \mathbf{y}\|_2^2 \;=\; \sum_{i=1}^{n} (\mathbf{u}_i(0) - y_i)^2 \;\le\; \sum_{i=1}^{n} \big(2\mathbf{u}_i(0)^2 + 2y_i^2\big) \;\le\; 8n\log(12n/\delta) + 2n \;\le\; C' n \log(n/\delta),$$

for a sufficiently large universal $C' > 0$ (absorbing the additive constants). This completes the proof. □

4 Proof of Lemma 2.3

Proof of Lemma 2.3. The proof has three steps. We first define, for each input $i \in [n]$, the random set $S_i \subseteq [m]$ of neurons whose ReLU activation pattern can flip under a perturbation of size $\le R$; we then bound $|S_i|$ in high probability using Gaussian small-ball; finally we convert this to an operator-norm bound on $\mathbf{H}(\mathbf{W}) - \mathbf{H}(\mathbf{W}(0))$.

Step 1: identifying flippable neurons. For each $i \in [n]$ and $r \in [m]$, define the event

$$A_{i,r}(R) \;\coloneqq\; \big\{\, \exists\, \mathbf{w} \in \mathbb{R}^d \text{ with } \|\mathbf{w} - \mathbf{w}_r(0)\|_2 \le R \text{ and } \mathbf{1}[\mathbf{w}^\top \mathbf{x}_i \ge 0] \ne \mathbf{1}[\mathbf{w}_r(0)^\top \mathbf{x}_i \ge 0] \,\big\}. \tag{13}$$

On the complement $A_{i,r}(R)^c$, every $\mathbf{w}$ within distance R of $\mathbf{w}_r(0)$ produces the same ReLU pattern as $\mathbf{w}_r(0)$ on input $\mathbf{x}_i$. The key set-containment is

$$A_{i,r}(R) \;\subseteq\; \big\{\, |\mathbf{w}_r(0)^\top \mathbf{x}_i| \le R \,\big\}. \tag{14}$$

Indeed, if there exists $\mathbf{w}$ with $\|\mathbf{w} - \mathbf{w}_r(0)\|_2 \le R$ flipping the sign of $\mathbf{w}^\top \mathbf{x}_i$ relative to $\mathbf{w}_r(0)^\top \mathbf{x}_i$, then by Cauchy-Schwarz $|\mathbf{w}^\top \mathbf{x}_i - \mathbf{w}_r(0)^\top \mathbf{x}_i| \le R$ (using $\|\mathbf{x}_i\| = 1$, Assumption 1.1), while $\mathbf{w}^\top \mathbf{x}_i$ and $\mathbf{w}_r(0)^\top \mathbf{x}_i$ have opposite signs, so the segment between them contains 0, forcing $|\mathbf{w}_r(0)^\top \mathbf{x}_i| \le R$.

Step 2: bounding the expected count. By Eq. (14) and Fact 1.11 applied to $\mathbf{w}_r(0)^\top \mathbf{x}_i \sim \mathcal{N}(0,1)$ (which uses $\|\mathbf{x}_i\| = 1$ from Assumption 1.1),

$$\mathbb{P}[A_{i,r}(R)] \;\le\; \mathbb{P}\big[|\mathbf{w}_r(0)^\top \mathbf{x}_i| \le R\big] \;\le\; R. \tag{15}$$

Let $S_i \coloneqq \{r \in [m] : A_{i,r}(R) \text{ occurs}\}$ and $|S_i|$ its cardinality. Then $\mathbb{E}|S_i| \le mR$ by linearity, since the events $A_{i,r}(R)$ are iid across r. By Markov, $|S_i| \le 3mR$ with probability $\ge 2/3$. To get a high-probability bound at level $\delta/(3n)$, we use a stronger concentration. Each $\mathbf{1}[A_{i,r}(R)]$ is a Bernoulli random variable bounded in $[0,1]$, so by Hoeffding's inequality (Fact 1.10, applied to the centered indicators $\mathbf{1}[A_{i,r}(R)] - \mathbb{P}[A_{i,r}(R)]$ with bound $b = 1$ and threshold $t = R$),

$$\mathbb{P}\Big[\, \tfrac{1}{m}|S_i| \;\ge\; 2R \,\Big] \;\le\; \mathbb{P}\Big[\, \tfrac{1}{m}\textstyle\sum_{r=1}^{m} \big(\mathbf{1}[A_{i,r}(R)] - \mathbb{P}[A_{i,r}(R)]\big) \;\ge\; R \,\Big] \;\le\; \exp\Big(-\tfrac{mR^2}{2}\Big),$$

where the first inequality uses $\mathbb{P}[A_{i,r}(R)] \le R$ from Eq. (15) to inflate the event: $\{\frac{1}{m}|S_i| \ge 2R\} \subseteq \{\frac{1}{m}\sum_r(\mathbf{1}[A_{i,r}(R)] - \mathbb{P}[A_{i,r}(R)]) \ge 2R - R = R\}$; the second applies one-sided Hoeffding (Fact 1.10 restricted to one tail) to the centered Bernoullis with $b = 1$. The right-hand side is $\le \delta/(3n)$ as soon as $mR^2 \ge 2\log(3n/\delta)$. Union-bounding over $i \in [n]$, with probability $\ge 1 - \delta/3$,

$$|S_i| \;\le\; 2mR \qquad \text{for every } i \in [n]. \tag{16}$$

Step 3: from neuron flips to operator-norm. Fix any $\mathbf{W}$ with $\|\mathbf{w}_r - \mathbf{w}_r(0)\|_2 \le R$ for every r, and any $i, j \in [n]$. Comparing Eq. (5) at $\mathbf{W}$ and at $\mathbf{W}(0)$, term by term:

$$\mathbf{H}(\mathbf{W})_{ij} - \mathbf{H}(\mathbf{W}(0))_{ij} \;=\; \frac{\mathbf{x}_i^\top \mathbf{x}_j}{m} \sum_{r=1}^{m} \Big(\mathbf{1}[\mathbf{w}_r^\top \mathbf{x}_i \ge 0]\mathbf{1}[\mathbf{w}_r^\top \mathbf{x}_j \ge 0] - \mathbf{1}[\mathbf{w}_r(0)^\top \mathbf{x}_i \ge 0]\mathbf{1}[\mathbf{w}_r(0)^\top \mathbf{x}_j \ge 0]\Big).$$

For any r such that neither $A_{i,r}(R)$ nor $A_{j,r}(R)$ occurs, the two activation patterns coincide and the summand vanishes. Hence only $r \in S_i \cup S_j$ contributes, and each such contribution is bounded by $|\mathbf{x}_i^\top \mathbf{x}_j|/m \le 1/m$ by Cauchy-Schwarz and Assumption 1.1. Thus

$$\big|\mathbf{H}(\mathbf{W})_{ij} - \mathbf{H}(\mathbf{W}(0))_{ij}\big| \;\le\; \frac{|S_i \cup S_j|}{m} \;\le\; \frac{|S_i| + |S_j|}{m}. \tag{17}$$

On the event of Eq. (16), the right-hand side is at most $4R$ for every i, j.

Finally, by $\|\mathbf{A}\|_{\mathrm{op}} \le n \max_{i,j} |\mathbf{A}_{ij}|$ (the same bound used in the proof of Lemma 2.2),

$$\|\mathbf{H}(\mathbf{W}) - \mathbf{H}(\mathbf{W}(0))\|_{\mathrm{op}} \;\le\; n \cdot 4R \;\le\; n \cdot 4 \cdot \tfrac{c\lambda_0}{n^2} \;=\; \tfrac{4c\lambda_0}{n} \;\le\; \tfrac{\lambda_0}{4},$$

where the second inequality applies the lemma's hypothesis $R \le c\lambda_0/n^2$, and the last inequality uses any $c \le 1/16$ together with $n \ge 1$. The conclusion holds simultaneously for every admissible $\mathbf{W}$ on the high-probability event of Eq. (16) (which depends only on $\mathbf{W}(0)$, not on $\mathbf{W}$). This completes the proof. □

Remark 4.1. The high-probability event in Lemma 2.3 is a property of the initialization $\mathbf{W}(0)$ alone (it asserts a uniform bound on $|S_i|$ for every i), so the conclusion holds simultaneously for every $\mathbf{W}$ in the R-ball, with no further union bound over $\mathbf{W}$. This is what enables the contradiction argument in Section 6.

5 Proof of Lemma 2.4

Proof of Lemma 2.4. The proof is by induction on s. The base case $s = 0$ holds trivially: $\|\mathbf{u}(0) - \mathbf{y}\|_2^2 \le (1 - \eta\lambda_0/2)^0 \|\mathbf{u}(0) - \mathbf{y}\|_2^2$ and $\mathbf{w}_r(0) = \mathbf{w}_r(0)$.

Fix $0 \le s \le t$ and assume the conclusion holds at every step $s' \le s$. We prove it at step $s + 1$. Throughout, we suppress arguments of $\mathbf{H}$ when the iterate $\mathbf{W}(s)$ is clear and write $\mathbf{H}_s := \mathbf{H}(\mathbf{W}(s))$.

Step 1: residual update. For each $i \in [n]$, by Eq. (3) and a first-order expansion of the piecewise-linear σ (where the "remainder" $e_i(s)$ captures only sign-flip contributions, not a quadratic Taylor error, since σ is affine on each open halfspace),

$$\mathbf{u}_i(s+1) - y_i \;=\; \mathbf{u}_i(s) - y_i \;-\; \eta \sum_{r=1}^{m} \langle \nabla_{\mathbf{w}_r} \mathbf{u}_i(\mathbf{W}(s)), \nabla_{\mathbf{w}_r} L(\mathbf{W}(s)) \rangle \;+\; e_i(s), \tag{18}$$

where $e_i(s)$ is the higher-order remainder. Plugging in Eq. (4) and the identity (Remark 1.6)

$$\sum_{r=1}^{m} \langle \nabla_{\mathbf{w}_r} \mathbf{u}_i(\mathbf{W}(s)), \nabla_{\mathbf{w}_r} \mathbf{u}_j(\mathbf{W}(s)) \rangle \;=\; \mathbf{H}_{sij}, \tag{19}$$

we obtain $\sum_r \langle \nabla_{\mathbf{w}_r} \mathbf{u}_i(\mathbf{W}(s)), \nabla_{\mathbf{w}_r} L(\mathbf{W}(s)) \rangle = \sum_j (\mathbf{u}_j(s) - y_j)\,(\mathbf{H}_s)_{ij}$, hence vectorizing

$$\mathbf{u}(s+1) - \mathbf{y} \;=\; \big(\mathbf{I} - \eta \mathbf{H}_s\big)(\mathbf{u}(s) - \mathbf{y}) \;+\; \mathbf{e}(s), \tag{20}$$

with $\mathbf{e}(s) = (e_1(s), \ldots, e_n(s))^\top$.

Step 2: bounding the higher-order remainder. We show $\|\mathbf{e}(s)\|_2 \le \frac{\eta\lambda_0}{4} \|\mathbf{u}(s) - \mathbf{y}\|_2$ for C_2 sufficiently small. By definition,

$$e_i(s) \;=\; \tfrac{1}{\sqrt{m}} \sum_{r=1}^{m} a_r \big(\sigma(\mathbf{w}_r(s+1)^\top \mathbf{x}_i) - \sigma(\mathbf{w}_r(s)^\top \mathbf{x}_i) - \sigma'(\mathbf{w}_r(s)^\top \mathbf{x}_i)\,(\mathbf{w}_r(s+1) - \mathbf{w}_r(s))^\top \mathbf{x}_i\big).$$

Now σ is 1-Lipschitz and the higher-order remainder $\sigma(z+\Delta) - \sigma(z) - \sigma'(z)\Delta$ vanishes whenever the activation pattern does not flip between z and $z+\Delta$, and is otherwise bounded by $|\Delta|$. Let $T_i(s) \subseteq [m]$ denote the set of neurons whose activation pattern flips on input $\mathbf{x}_i$ at step s, i.e., $\mathbf{1}[\mathbf{w}_r(s+1)^\top \mathbf{x}_i \ge 0] \ne \mathbf{1}[\mathbf{w}_r(s)^\top \mathbf{x}_i \ge 0]$. Then

$$|e_i(s)| \;\le\; \frac{1}{\sqrt{m}} \sum_{r \in T_i(s)} |a_r| \cdot |(\mathbf{w}_r(s+1) - \mathbf{w}_r(s))^\top \mathbf{x}_i| \;\le\; \frac{1}{\sqrt{m}} \sum_{r \in T_i(s)} \|\mathbf{w}_r(s+1) - \mathbf{w}_r(s)\|_2 \,. \tag{21}$$

Each per-neuron gradient is bounded by $\sqrt{n}\,\|\mathbf{u}(s) - \mathbf{y}\|_2 / \sqrt{m}$ (the standard ReLU gradient bound; see Eq. (4), $\|\mathbf{x}_i\| = 1$, $|a_r| = 1$, and Cauchy-Schwarz on $\sum_i (\mathbf{u}_i(s) - y_i)$):

$$\|\mathbf{w}_r(s+1) - \mathbf{w}_r(s)\|_2 \;=\; \eta\, \|\nabla_{\mathbf{w}_r} L(\mathbf{W}(s))\|_2 \;\le\; \frac{\eta\sqrt{n}}{\sqrt{m}}\, \|\mathbf{u}(s) - \mathbf{y}\|_2 \,. \tag{22}$$

Substituting in Eq. (21),

$$|e_i(s)| \;\le\; \frac{|T_i(s)|}{\sqrt{m}} \cdot \frac{\eta\sqrt{n}}{\sqrt{m}}\, \|\mathbf{u}(s) - \mathbf{y}\|_2 \;=\; \frac{\eta\sqrt{n}\,|T_i(s)|}{m}\, \|\mathbf{u}(s) - \mathbf{y}\|_2 \,. \tag{23}$$

We claim $T_i(s) \subseteq S_i$. Indeed, by the third hypothesis of the lemma, both $\mathbf{w}_r(s)$ and $\mathbf{w}_r(s+1)$ are within R of $\mathbf{w}_r(0)$; if $r \in T_i(s)$, i.e., $\mathbf{1}[\mathbf{w}_r(s)^\top \mathbf{x}_i \ge 0] \ne \mathbf{1}[\mathbf{w}_r(s+1)^\top \mathbf{x}_i \ge 0]$, then the existential clause defining $A_{i,r}(R)$ in Eq. (13) is witnessed by the pair $(\mathbf{w}_r(s), \mathbf{w}_r(s+1))$ (with one of the two playing the role of the reference $\mathbf{w}_r(0)$): more precisely, both vectors lie in the R-ball around $\mathbf{w}_r(0)$, so the sign-flip between them forces $\mathbf{w}_r(0)^\top \mathbf{x}_i$ to lie in the segment connecting $\mathbf{w}_r(s)^\top \mathbf{x}_i$ and $\mathbf{w}_r(s+1)^\top \mathbf{x}_i$, which straddles 0. This means $A_{i,r}(R)$ occurs (the pair $(\mathbf{w}_r(0), \mathbf{w}_r(s))$ already witnesses it if $\mathbf{w}_r(s)$ has the same sign as $\mathbf{w}_r(s+1)$, and similarly with $\mathbf{w}_r(s+1)$ otherwise; in either case the witness is within R of $\mathbf{w}_r(0)$). Hence $r \in S_i$. Combined with the second hypothesis $|S_i| \le 2mR$,

$$|T_i(s)| \;\le\; |S_i| \;\le\; 2mR. \tag{24}$$

Plugging $|T_i(s)| \le 2mR$ into Eq. (23), then summing,

$$\|\mathbf{e}(s)\|_2 \;\le\; \sqrt{n} \cdot \max_i |e_i(s)| \;\le\; \sqrt{n} \cdot 2\eta\sqrt{n}\, R\, \|\mathbf{u}(s) - \mathbf{y}\|_2 \;=\; 2\eta n R\, \|\mathbf{u}(s) - \mathbf{y}\|_2 \,. \tag{25}$$

With $R = O(\lambda_0/n^2)$ from the perturbation hypothesis of Lemma 2.3, the prefactor in Eq. (25) is at most $\frac{\eta\lambda_0}{4}$ for c in $R \le c\lambda_0/n^2$ sufficiently small.

Step 3: contraction of the residual. Combining Eq. (20) with the triangle inequality and Eq. (25):

$$\|\mathbf{u}(s+1)-\mathbf{y}\|_2^2 \;\le\; \big(\,\|\mathbf{I}-\eta\mathbf{H}_s\|_{\mathrm{op}} + \tfrac{\eta\lambda_0}{4}\big)^2\,\|\mathbf{u}(s)-\mathbf{y}\|_2^2\,. \tag{26}$$

By hypothesis, $\lambda_{\min}(\mathbf{H}_s) \ge \lambda_0/2$, and trivially $\|\mathbf{H}_s\|_{\mathrm{op}} \le n$ since each entry of $\mathbf{H}_s$ is in $[0,1]$ by Eq. (5) and Cauchy-Schwarz (so $\|\mathbf{H}_s\|_F \le n$ implies $\|\mathbf{H}_s\|_{\mathrm{op}} \le n$). With $\eta = C_2\lambda_0/n^2$ and C_2 small enough that $\eta n \le 1$, we have $\|\mathbf{I}-\eta\mathbf{H}_s\|_{\mathrm{op}} \le 1-\eta\lambda_0/2$. Hence

$$\|\mathbf{u}(s+1)-\mathbf{y}\|_2^2 \;\le\; \big(1-\tfrac{\eta\lambda_0}{4}\big)^2\,\|\mathbf{u}(s)-\mathbf{y}\|_2^2 \;\le\; \big(1-\tfrac{\eta\lambda_0}{2}\big)\,\|\mathbf{u}(s)-\mathbf{y}\|_2^2\,.$$

Combined with the inductive hypothesis, this gives $\|\mathbf{u}(s+1)-\mathbf{y}\|_2^2 \le (1-\eta\lambda_0/2)^{s+1}\,\|\mathbf{u}(0)-\mathbf{y}\|_2^2$.

Step 4: distance from initialization. By Eq. (22) and the inductive contraction just established,

$$\|\mathbf{w}_r(s+1)-\mathbf{w}_r(0)\|_2 \;\le\; \sum_{s'=0}^{s}\big\|\mathbf{w}_r(s'+1)-\mathbf{w}_r(s')\big\|_2 \;\le\; \sum_{s'=0}^{s}\frac{\eta\sqrt{n}}{\sqrt{m}}\,\big\|\mathbf{u}(s')-\mathbf{y}\big\|_2\,.$$

Using $\|\mathbf{u}(s')-\mathbf{y}\|_2 \le (1-\eta\lambda_0/2)^{s'/2}\,\|\mathbf{u}(0)-\mathbf{y}\|_2$ and summing the geometric series with rate $\sqrt{1-\eta\lambda_0/2} \le 1-\eta\lambda_0/4$,

$$\|\mathbf{w}_r(s+1)-\mathbf{w}_r(0)\|_2 \;\le\; \frac{\eta\sqrt{n}}{\sqrt{m}}\,\|\mathbf{u}(0)-\mathbf{y}\|_2\cdot\frac{1}{1-(1-\eta\lambda_0/4)} \;=\; \frac{4\sqrt{n}\,\|\mathbf{u}(0)-\mathbf{y}\|_2}{\sqrt{m}\,\lambda_0}.$$

This is the claimed bound. The induction step is complete, and so the lemma holds. □

6 Proof of Theorem 2.1

Proof of Theorem 2.1. The proof combines the three structural lemmas Lemmas 2.2 to 2.4 and the initial-residual control Lemma 2.5 through a fixed-point / contradiction argument. We condition on the intersection of three high-probability events, then use Lemma 2.4 as the inductive engine in a stay-in-ball regime that we maintain by contradiction.

Step 1: high-probability events. Let $R \coloneqq 4\sqrt{n}\,\|\mathbf{u}(0)-\mathbf{y}\|_2/(\sqrt{m}\lambda_0)$ as in Eq. (9). Define

$$\begin{aligned}
\mathcal{E}_1 &\coloneqq \big\{\,\|\mathbf{H}(\mathbf{W}(0))-\mathbf{H}^\infty\|_{\mathrm{op}} \le \lambda_0/4\big\},\\
\mathcal{E}_2 &\coloneqq \big\{\,|S_i| \le 2mR \;\text{ for every } i\in[n]\,\big\},\\
\mathcal{E}_3 &\coloneqq \big\{\,\|\mathbf{u}(0)-\mathbf{y}\|_2^2 \le C'n\log(n/\delta)\big\},
\end{aligned}$$

where S_i is the random set defined in Eq. (13) (which depends on $\mathbf{W}(0)$ and the radius R), and $\mathcal{E} \coloneqq \mathcal{E}_1\cap\mathcal{E}_2\cap\mathcal{E}_3$. By Lemma 2.2 ($\mathbb{P}[\mathcal{E}_1^c] \le \delta/3$), the proof of Lemma 2.3 at radius R (Eq. (16); $\mathbb{P}[\mathcal{E}_2^c] \le \delta/3$, provided $R \le c\lambda_0/n^2$ — verified below), Lemma 2.5 ($\mathbb{P}[\mathcal{E}_3^c] \le \delta/3$), and a union bound,

$$\mathbb{P}[\mathcal{E}] \;\ge\; 1-\delta.$$

Step 2: verifying the perturbation-radius hypothesis. We must check that the radius R implied by Lemma 2.4 satisfies $R \le c\lambda_0/n^2$ as required by Lemma 2.3. On $\mathcal{E}_3$,

$$R \;=\; \frac{4\sqrt{n}\;\|\mathbf{u}(0)-\mathbf{y}\|_2}{\sqrt{m}\,\lambda_0} \;\le\; \frac{4\sqrt{n}\cdot\sqrt{C' n\log(n/\delta)}}{\sqrt{m}\,\lambda_0} \;=\; \frac{4\sqrt{C'}\,n\sqrt{\log(n/\delta)}}{\sqrt{m}\,\lambda_0}. \tag{27}$$

The condition $R \le c\lambda_0/n^2$ becomes

$$\frac{4\sqrt{C'}\,n\sqrt{\log(n/\delta)}}{\sqrt{m}\,\lambda_0} \;\le\; \frac{c\lambda_0}{n^2} \quad\Longleftrightarrow\quad m \;\ge\; \frac{16C'\,n^6\log(n/\delta)}{c^2\lambda_0^4}.$$

This is implied by the theorem's width hypothesis $m \ge C_1 n^6/(\lambda_0^4\delta^3)$ for C_1 sufficiently large (using $\log(n/\delta) \le \mathrm{poly}(1/\delta)$). With this width, all three events $\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3$ are guaranteed simultaneously by Lemmas 2.2, 2.3 and 2.5 (which each require $m \ge Cn^2\log(n/\delta)/\lambda_0^2$, dominated by the above), and the union bound gives $\mathbb{P}[\mathcal{E}] \ge 1-\delta$.

Step 3: inductive stay-in-ball argument by contradiction. We claim that on $\mathcal{E}$, for every $t \ge 0$,

$$(\star_t) \qquad \|\mathbf{u}(t)-\mathbf{y}\|_2^2 \le \left(1-\tfrac{\eta\lambda_0}{2}\right)^t \|\mathbf{u}(0)-\mathbf{y}\|_2^2 \quad\text{and}\quad \|\mathbf{w}_r(t)-\mathbf{w}_r(0)\|_2 \le R \;\text{ for all } r. \tag{28}$$

$(\star_0)$ is immediate. Suppose $(\star_t)$ holds for some $t \ge 0$; we prove $(\star_{t+1})$.

By $(\star_t)$, every iterate $\mathbf{W}(0), \mathbf{W}(1), \ldots, \mathbf{W}(t)$ has each column within R of the corresponding initial column. Hence Eq. (17) of Lemma 2.3's proof — which derives the operator-norm bound from the flip-count bound — applies to each iterate $\mathbf{W}(s)$, $s \in \{0,1,\ldots,t\}$, on $\mathcal{E}_2$ (this is where the *uniform* perturbation bound matters — see Remark 4.1). Combined with the conversion from entrywise to operator norm,

$$\|\mathbf{H}(\mathbf{W}(s)) - \mathbf{H}(\mathbf{W}(0))\|_{\mathrm{op}} \;\le\; \tfrac{\lambda_0}{4} \qquad \text{for all } s \in \{0,1,\ldots,t\}.$$

Combining with $\mathcal{E}_1$, the triangle inequality, and Weyl's inequality,

$$\lambda_{\min}(\mathbf{H}(\mathbf{W}(s))) \;\ge\; \lambda_{\min}(\mathbf{H}^\infty) - \|\mathbf{H}(\mathbf{W}(s)) - \mathbf{H}^\infty\|_{\mathrm{op}} \;\ge\; \lambda_0 - \tfrac{\lambda_0}{4} - \tfrac{\lambda_0}{4} \;=\; \tfrac{\lambda_0}{2}, \tag{29}$$

for every $s \in \{0,1,\ldots,t\}$. The hypotheses of Lemma 2.4 are therefore met up to step t: (i) the spectral lower bound is just Eq. (29); (ii) on $\mathcal{E}_2$, the flip-count bound $|S_i| \le 2mR$ holds by the proof of Lemma 2.3 (Eq. (16), which $\mathcal{E}_2$ subsumes since $\mathcal{E}_2$ is the conclusion of Lemma 2.3); (iii) the perturbation $\|\mathbf{w}_r(s)-\mathbf{w}_r(0)\|_2 \le R$ for $s \le t$ is exactly the second clause of the inductive hypothesis $(\star_t)$; (iv) the constants c, C_2 requirements are imposed once at the theorem level. Therefore Lemma 2.4 applied at this t delivers both conclusions: $\|\mathbf{u}(t+1)-\mathbf{y}\|_2^2 \le (1-\eta\lambda_0/2)^{t+1}\|\mathbf{u}(0)-\mathbf{y}\|_2^2$ and $\|\mathbf{w}_r(t+1)-\mathbf{w}_r(0)\|_2 \le 4\sqrt{n}\,\|\mathbf{u}(0)-\mathbf{y}\|_2/(\sqrt{m}\lambda_0) = R$ for every r. This is $(\star_{t+1})$.

By induction, $(\star_t)$ holds for every $t \ge 0$, which proves Eq. (8) and Eq. (9). The conclusion holds with probability $\ge 1-\delta$ over the random initialization (Assumption 1.3). This completes the proof. □

Remark 6.1 (On the contradiction phrasing). The argument above is conventionally phrased "by contradiction": let $T^* := \inf\{t : \exists r, \|\mathbf{w}_r(t)-\mathbf{w}_r(0)\|_2 > R\}$; if $T^* < \infty$ then Lemma 2.4 applied at $t = T^*-1$ shows $\|\mathbf{w}_r(T^*)-\mathbf{w}_r(0)\|_2 \le R$, contradicting the definition of T^*. The induction $(\star_t)$ in Step 3 is the same content. We use induction directly because it is cleaner; the contradiction phrasing is recovered by reading the induction step's conclusion as "the iterate cannot first breach the ball at time $t+1$".

Remark 6.2 (Reconciliation with the headline width). The width condition $m \geq C_1 n^6/(\lambda_0^4 \delta^3)$ in Eq. (7) dominates the three subsidiary conditions appearing in the lemma hypotheses:

- $m \geq Cn^2 \log(n/\delta)/\lambda_0^2$ from Lemma 2.2;
- $m \geq Cn^2 \log(n/\delta)/\lambda_0^2$ from Lemma 2.3 (with R as in Eq. (27));
- $m \geq 16C' n^6 \log(n/\delta)/(c^2 \lambda_0^4)$ from the requirement that $R \leq c\lambda_0/n^2$.

The third dominates and matches $m \gtrsim n^6/(\lambda_0^4 \delta^3)$ up to $\mathrm{polylog}(n, 1/\delta)$ factors absorbed into C_1. This n^6-polynomial in m is the same scaling appearing in [DZPS19]; tighter polynomials are available in follow-up work but lie outside the scope of this proof.

References

[DZPS19] Simon S. Du, Xiyu Zhai, Barnabás Póczos, and Aarti Singh. Gradient descent provably optimizes over-parameterized neural networks. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1810.02054.